

Harsh Kumar Jha

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SUMMARY

Engineering student focused on AI products, backend systems, and LLM safety tooling. Built projects in hallucination detection, prompt-risk filtering, and voice-enabled ML systems. Strong in rapid prototyping, shipping products, and learning quickly in startup environments.

SKILLS

Languages: Python, SQL, Java, C++

Backend: FastAPI, REST APIs, JSON Workflows, Uvicorn, Deployment Basics

AI / ML: scikit-learn, Pandas, NumPy, Matplotlib, Model Integration, Prompt Engineering

LLM Systems: Hallucination Detection, Prompt Injection Detection, RAG Validation, Safety Prototyping

Tools: Git, GitHub, Jupyter, Google Colab, Postman, Docker Basics

EXPERIENCE

Research & Development Intern — MathonGo (Remote)

Jun 2025 – Sep 2025

- Developed and validated 2000+ Physics problems and structured question-bank content for JEE/NEET programs.
- Performed formula verification, logic validation, and concept consistency checks.
- Used AI-assisted workflows to improve review speed and output quality.
- Managed tagged datasets for retrieval efficiency and syllabus mapping.
- Worked in a remote deadline-driven production workflow.

PROJECTS

Hallucination Hunter (GitHub)

Python, FastAPI

- Built LLM hallucination detection system to compare generated outputs with source context.
- Used pretrained NLI and semantic similarity models for contradiction / unsupported claim checks.
- Implemented claim-level analysis, confidence summaries, and explainable outputs.
- Developed FastAPI backend for RAG and AI evaluation workflows.

Prompt Police (GitHub)

Python, FastAPI

- Built prompt-risk detection system for jailbreak and prompt-injection style attacks.
- Combined TF-IDF scoring, heuristic filters, and open-source model workflows.
- Created REST API safety layer for screening prompts before LLM execution.
- Focused on low-latency inference and deployment readiness.

AgriMind (GitHub)

Python, FastAPI

- Built voice-enabled crop recommendation system using ML prediction workflow.
- Used RandomForest model on agriculture datasets for crop suggestion logic.
- Integrated Whisper speech-to-text and text-to-speech interaction flow.
- Developed backend APIs for real-time prediction and voice processing.

OPEN SOURCE

Contributor — NSOC & GSSOC

2025

- Contributed to repositories through issue resolution, documentation updates, and feature improvements.
- Worked with Git/GitHub workflows including forks, pull requests, and collaboration practices.
- Improved understanding of production codebases and open-source contribution standards.

EDUCATION

Maharaja Surajmal Institute of Technology, New Delhi

2025 – 2029

B.Tech, Electronics and Communication Engineering

CGPA: 8.14 / 10

CERTIFICATION

McKinsey.org Forward Program

2025